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THE IMPORTANCE OF BOUNDARY SPANNING ROLES IN STRATEGIC DECISION-MAKING [1]

DAVID B. JEMISON
Graduate School of Business, Stanford University

ABSTRACT

The influence of boundary spanning roles on strategic decision-making was explored in a field study in fifteen organizations. Results support the importance of boundary spanning roles in the strategic decision-making process and the relationship of technology to the differential strategic decision-making influence of different boundary spanning roles.

INTRODUCTION

THE study of the sources of influence on strategic decision-making is an important area of research in strategic management. In this regard a fruitful avenue of inquiry is that of interorganizational relations and the boundary spanning units that link the organization with its environment. Boundary spanning units are important in strategic decision-making because of their ability to recognize and deal with trends or changes in the environment – an important characteristic of complex organizations that wish to survive.

Boundary spanning and interorganizational influence on strategic decision-making have usually been addressed separately in research contexts. Although it is acknowledged that boundary spanning leads to influence on strategic decisions (Aldrich and Herker, 1977; Kochan, 1975; Leifer and Delbecq, 1978; Thompson, 1967), there has been little empirical exploration of this relationship. This paper reports the results of a field study in fifteen organizations representing three industries that addressed three primary gaps in our knowledge regarding the relationship between boundary spanning and strategic decision-making influence.

1. The study of the influence of different spanning roles on strategic decisions.

Earlier studies of boundary spanning influences on strategic decisions have simply maintained that, in general, boundary spanners participate more in strategic decisions (Aiken and Hage, 1972; Filley and Grimes, 1967; Leifer, 1975; Pettigrew, 1972; Strauss, 1962). These studies did not explore the

Address for reprints: Professor D. B. Jemison, Graduate School of Business, Stanford University, Stanford, CA. 94305, U.S.A.

relative influence of different boundary spanning roles on strategic decisions. This study tests hypotheses regarding the relative influence of three separate, empirically-derived boundary spanning roles.

2. The study of the conditions under which different boundary spanning roles' influence on strategic decisions may vary.

Different contingencies may cause a variation in influence among boundary spanning roles (Aldrich and Herker, 1977; Miles, 1980). However, no studies have explored the conditions under which the relative influence of different boundary roles on the strategic decisions of the organization will change. The Aston group explained differential subunit power on the basis of the subunit's ability to deal with strategic contingencies facing the organization (Hickson *et al.*, 1971; Hinings *et al.*, 1974). Although they speak of a 'strategic' contingencies theory, their research orientation was directed to operations management of essentially manufacturing firms. In contrast, this study explored the relative influence of boundary spanning roles on strategic decisions in a variety of different organization types.

3. The study of boundary spanners' influence on strategic decisions at the level of the organization's senior decision-makers.

Previous studies that have addressed boundary spanners' influence on strategic decisions have also suffered from inadequate definitions of strategic decisions (Aiken and Hage, 1972; Bacharach and Aiken, 1976) or an organizational level of analysis substantially subordinate to that of the senior decision-making executives (Bacharach and Aiken, 1976; Leifer, 1975; Spekman, 1979). This study examines both strategic decisions and strategic decision-makers from the organization-wide perspective of the enterprises' senior executives.

CONCEPTUAL BACKGROUND AND HYPOTHESES

Influence on Strategic Decision-Making

Every organization, to one degree or another, develops a strategy that coaligns its resources and competencies with its environment. The development and maintenance of this strategy is a key determinant of the continued success and prosperity of the organization. In this light, strategic decisions are seen as those decisions that profoundly affect the future success and destiny of the organization by assuring that it is adequately responsive to environmental requirements.

The concept of influence, as studied here, refers to the ability of a department (or other subunit) to affect the outcome of a particular strategic decision for its organization. Dahl (1963) suggests that to meaningfully consider the influence of an individual or a group, one must first identify the domain, scope and means of that influence. For purposes of this study, the domain of a subunit's (department's) influence is each of the other subunits of the organization. The scope of its influence are the strategic decisions facing the organization. The means (sources) of the subunit's influence are the *raison d'etre* for the hypotheses tested in this paper.

Influence (the ability to affect the outcome of decisions) is seen to accrue to individuals or subunits in the organization because the others become dependent upon them (Blau, 1964; Emerson, 1962; Hickson *et al.*, 1971; Thompson, 1967). This dependence arises because the group with influence is able to deal with important matters more effectively than other groups. In the context of this study, a specific department's influence on strategic decisions will be associated with its ability to ameliorate strategic contingencies for the organization. Strategic contingencies are seen as decision variables that arise from changes in factors which materially affect the organization's ability to survive and prosper but over which managers have relatively little control. The process of boundary spanning (interacting with others outside the organization) gives certain departments the ability to deal with these strategic contingencies. Hambrick (1981a) found that managers who coped with their industry's dominant environmental requirements (strategic contingencies) had high power in strategic decisions. This paper explores specific ways in which departments are able to create dependencies with their boundary spanning activities.

Boundary Spanning and Influence on Strategic Decisions

Every organization faces a unique environment that continually presents challenges that affect its ability to survive and prosper. Although the nature of environmental challenges is different for each organization, some similarities exist in the ways in which the environment is dealt with. Specifically, organizations establish boundary spanning units to manage the environmental transactions necessary for success (Thompson, 1967).

This study views boundary spanning as the set of activities involved with organization-environment interaction. Any activity that links an organization with its task environment can be seen as boundary spanning behaviour. A variety of boundary spanning roles have been postulated by theorists. The most predominate themes suggest that boundary spanning roles are involved with either inputs to the organization or outputs from the organization of information or resources. This corresponds with Adams' (1976) boundary spanning roles of acquisition and disposal. But, these roles are too general for the more detailed study that interorganizational research demands.

Others have attempted a theoretical identification of the sorts of boundary roles present in organizations. Katz and Kahn (1966) see three boundary roles: (1) procuring resources and disposing of outputs, (2) relating the organization to its larger community or social system, and (3) adapting the organization to the future by gathering information about trends and planning to meet these developments. Aiken and Hage (1972) see boundary spanning roles as those roles that link the focal organization with other organizations or social systems and are directly relevant for the goal attainment of the focal organization. In a similarly general view, Leifer and Delbecq (1978) see the function of boundaries (and therefore the boundary spanning activities of the organization) as protecting the organization from environmental stress and acting as regulators of information and material flow between the organization and its environment. Aldrich and Herker (1977) propose two primary classes of boundary spanning functions: information processing and external representation.

Despite these theoretical conjectures, there is a dearth of empirical evidence identifying actual boundary spanning roles in organizations. Keller and Holland (1975) identified information processing boundary spanning roles in a research and development organization. Miles (1976) also studied a research and development organization and found three boundary spanning roles present: (1) linking and coordinating, (2) information filtration and transfer, and (3) feedback collection and dissemination.

Earlier theoretical and empirical classifications of boundary spanning roles were not appropriate for this study. The theoretical roles were not sufficiently precise for empirical research purposes. In addition, the empirical classifications were not useful for this study because they were developed to deal with boundary spanning at a level subordinate to that of the top management group or lacked generalizability because they were developed solely from the context of the R and D function.

To remedy this problem, an empirically-derived classification of boundary spanning roles was developed. Details of the development of this classification are presented in the methodology section of the paper. But for purposes of hypothesis development, the three boundary spanning roles used in this study are briefly described and presented here. They are: (1) Information acquisition and control, (2) Domain determination and interface and (3) Physical input control. The activities associated with these roles are presented below:

(1) *Information acquisition and control*
Acquire information needed by the organization from outside sources. Control the disposition of the information by deciding to whom, when and what portions of the acquired information should be given to others.

(2) *Domain determination and interface*
Decide on the kinds of customers the organization will pursue and the method by which the product will be provided to the customers. Meet with the customers and provide information to others to create a favourable image of the organization.

(3) *Physical input control*
Decide on the kinds, quality and delivery schedules of physical inputs acquired from outside the organization (*e.g.* raw material, funds, personnel, supplies, *etc.*).

Table 1 relates the boundary-spanning roles used here with those previously developed by theoreticians.

Table I. Comparison of empirically derived boundary spanning roles with boundary spanning roles from the literature

	Empirically Derived Boundary Roles		
	Information acquisition and control	*Domain determination and interface*	*Physical input control*
Adams (1976)			
– Acquisition	X		X
– Disposal		X	
Aiken and Hage (1972)			
– Link focal organization with other organizations	X	X	X
Aldrich and Herker (1977)			
– Information processing	X		X
– External representation		X	X
Katz and Kahn (1966)			
– Procuring resources and disposing of outputs		X	X
– Relating the organization to the larger community		X	
– Adapting the organization to the future	X	X	
Leifer and Delbecq (1978)			
– Protect organization from environmental stress	X		X
– Regulate information flow between organization and environment	X	X	
Miles (1976)			
– Represent organization		X	
– Scar environment	X		
– Protect organization	X	X	X
– Process information	X		
– Transact to acquire inputs and dispose of outputs	X		X
– Link and coordinate	X	X	X

These three boundary spanning roles can be and are carried out by more than one organizational function. In fact, prior research has shown that boundary spanning roles are not synonymous with a particular position (Tushman and Scanlan, 1981) and that environmental scanning is highly cross-functional (Hambrick, 1981b). That is, a marketing person can acquire information about the changing environment, meet with a customer and enhance the organization's image – all in one sales call. Similarly, a corporate treasurer, when negotiating a credit line with a bank will not only get the needed funds but may also exchange intelligence information with contemporaries at the bank. Finally, it should also be noted that the same boundary spanning activities may be performed by different organizational functions. In the examples above, the information acquisition and control role was performed by both the marketing person and the treasurer.

Departments that engage in these boundary spanning roles will gain influence over strategic decisions to the extent that the boundary role is related to a critical contingency in that organization. This leads us to an important *a priori* assumption of this paper. The influence of a boundary spanning role on strategic decisions will be affected by the technology of the organization.

Technology is seen as an important determinant of the intraorganizational influence patterns because the division of labour arising from the organization's technology creates interdependencies from which influence on strategic decisions is derived (Thompson, 1967). As we discussed earlier, influence will accrue to the departments that are able to cope with the uncertainities caused by these interdependencies (Hickson *et al.*, 1971). Consequently, to the extent that departments cope with environmentally-related contingencies through boundary spanning, they will gain influence over strategic decisions.

The concept of technology as used in this study refers to the internal organizational processes (activities) that are needed to transform inputs into useful outputs. In the words of economists, an organization's technology is the composite of its internal processes by which utility is created for its customers or clients. Because there are many definitions available for the construct, a set of guidelines was needed to select the conceptualization of technology most appropriate for this study. Three criteria were used:

(1) It must be from the perspective of the total organization rather than a subunit. An examination of the manner in which influences on strategic decisions change under varying technologies requires this total organization perspective.
(2) It must be intuitively related to the concept of boundary spanning roles and their influences on strategic decisions, *i.e.* different technologies need different boundary spanning activities to survive and thus their influence on strategic decisions changes.
(3) It must readily differentiate among organizations for assignment into the appropriate category.

Such a typology of technology does not exist. But, from among the myriad of categorizations and definitions, Thompson's (1967) typology of long-linked, mediating and intensive technologies was chosen because it fulfills at least the first two conditions and, if operationalized properly, the third condition above. Other views of technology were considered but not used for several reasons. Some were only applicable in manufacturing organizations (Amber and Amber, 1962; Burns and Stalker, 1961; Hickson *et al.*, 1969; Khandwalla, 1974; Woodward, 1965). Conversely, others were too general for consistent assignment of organizations to the categories (Hunt, 1970; Perrow, 1970).

Mahoney and Frost (1974) and Hitt (1975) operationalized Thompson's typology on the basis of discretion available to managers. Their approach was not followed here because of the difficulty of applying the managerial discretion approach at a total organizational level. Instead, Thompson's typology was operationalized using the degree of interdependence and the types of coordinating mechanisms present. Long-linked technologies are serially interdependent and are coordinated with detailed plans for work flow. Mediating technologies have a pooled interdependence and use common standards to coordinate inputs and outputs. Finally, intensive technologies are reciprocally interdependent and require mutual adjustment for coordination among the parties. Three industries were selected to correspond to each of the technologies:

Food processors – Long-linked technology
Financial institutions – Mediating technology
General hospitals – Intensive technology

Hypotheses

This paper argues that the technology of an organization creates internal interdependencies which allow departments engaging in certain types of boundary spanning activities to gain influence over strategic decision-making. Three hypotheses provided direction in this regard:

Hypothesis 1

In long-linked technologies, the boundary spanning roles of Domain determination and interface and Physical input control will have more influence on strategic decisions than the boundary spanning role of Information acquisition and control.

Organizations with long-linked technologies have internal processes that are characterized by a high degree of sequential synchronization (*i.e.* they have serial interdependence). This serial interdependence requires careful planning to coordinate and sequence the work flow. Thus, the critical contingencies for these organizations would appear to be the ability to minimize disruptions to the work flow by sealing off the technical core of the organization from extraorganizational influence. This protection allows the internal processes to maintain their coordinated operations in the most sequentially efficient fashion possible.

The functions that are best suited to protect the technical core are those of input acquisition and output disposal (the boundary spanning roles of

Physical input control and Domain determination and interface). Thompson (1967) suggests that this protection can take place with strategies of buffering (*e.g.* developing buffer stocks) and smoothing (*e.g.* offering inducements to suppliers and customers to level the patterns of supply and demand). Landsberger (1961), Woodward (1965), Perrow (1970) and Pfeffer and Leblebici (1973) support this hypothesis when they suggest that in production organizations the marketing and sales people will be most influential because of the need to have an efficient throughput of large volumes of product.

Hypothesis 2

In mediating technologies, the boundary spanning roles of Domain determination and interface and Physical input control will have more influence on strategic decisions than the boundary spanning role of Information acquisition and control.

Mediating technologies face a slightly different situation. In mediating technologies the internal processes are highly standardized to insure equity among the organization's clients. For example, it is important that a financial institution treats each customer equitably with respect to rates on deposits and charges for loans. The standardization of the firm's internal processes allows this equitable treatment to take place. But high degrees of sequential synchronization are not important because these organizations act as pooling agents for various resources or assets provided by one group of clients and disposed of to another group of clients. These people provide slack resources and act as buffers that do away with the need for precise sequential synchronization. Thus, the critical contingencies are the maintenance of a sufficiently large pool of resources on which clients may draw and insuring the continuous utilization of those resources by clients. The boundary spanning roles of Physical input control and Domain determination and interface are in the best position to deal with these contingencies.

Hypothesis 3

In intensive technologies, the boundary spanning role of Information acquisition and control will have more influence on strategic decisions than either Domain determination and interface or Physical input control.

The internal processes of organizations with intensive technologies are characterized by a need for neither a high degree of sequential synchronization nor a high degree of standardization. These organizations serve their clients' needs on a case-by-case basis and an integral part of the process is a reciprocal information exchange.

The client of an organization with an intensive technology is usually brought into the organization before any services are performed. All relevant information is then obtained from the client with the result being a specific programme developed to suit the pecularities of the client's situation. The organization's resources needed to solve these problems can be directed only after the needed information is acquired. A programme of services (*e.g.* treatments) is then presented and after it has had an opportunity to take effect, more information is gathered and a mutual adjustment takes place. This suggests that the critical contingency for intensive technologies is the function of information acquisition – the boundary spanning role of Information acquisition and control. The work of Rosen (1976) and Pettigrew (1972) supports the influence of information gatekeepers in organizations.

METHODOLOGY

Sample Selection

Three industries were selected to represent the three types of technology used in this study. They are:

Long-linked technology – Food processors
Mediating technology – Financial institutions
Intensive technology – General hospitals

Food processors were selected to represent long-linked technologies because of the sequentially synchronized internal processes required to transform the raw food into a more edible or storable form. Because the raw materials are highly perishable and the period of availability of fresh foods is often short, processing facilities must have highly efficient, well articulated procedures and plans to clean, process, package, store and ship the food products. These processing activities are very serially interdependent and lend themselves to coordination by planning.

Financial institutions represent mediating technologies because their internal activities require coordination by standardization but do not need to be serially interdependent. It would be fiscally unsound and economically inefficient to match each depositor with a borrower. Consequently, the flow of funds from depositor to borrower is not necessarily highly sequentially synchronized. Instead, a pool of funds is maintained by the financial institution into which customers can deposit and from which other customers can borrow, the pooled interdependence of which Thompson (1967) writes. However, because the financial institution deals with a large number of borrowers and depositors, highly standardized procedures for deposits and loans are developed to insure fair treatment, maintain adequate control over funds, and promote economic efficiency in the funds transfer process.

Intensive technologies are represented by general hospitals. Before a patient can be diagnosed and treated, important information about his or her condition must be gathered. Neither of these processes, information gathering or treatment is necessarily highly sequentially synchronized nor highly standardized. Instead, the process is reciprocally interdependent where the outputs of the patient (data on his or her condition) and the physician (a prescribed treatment regimen) reciprocally become the inputs for each other.

Several criteria were used to select organizations for this study:

(1) The organizations had to fit into the operational definition of technology as presented above.
(2) To control for the confounding effects of size, the organizations had to be as homogeneous in size (with respect to technology type) as possible. There was a lower limit though – each organization has to be large enough to have clearly designated functions below the level of C.E.O.
(3) The organization's operations had to be concentrated in one line of business. The dependent variable, influence on strategic decision-making, is a concept that is slippery enough without adding confusion due to materially different product lines within the same organization.
(4) The organization had to be an autonomous, self-contained entity.

Fifteen organizations were selected for study, five each from the food processing (long-linked technology), financial institutions (mediating technology) and general hospital (intensive technology) industries.

The initial contact with the firm was in a personal interview conducted with the C.E.O. The purpose of the interview was to secure cooperation of the firm, make sure that the firm was properly categorized with respect to technology, learn about the strategic decisions facing the firm and determine which departments in the organization influenced the outcomes of those strategic decisions.

After each C.E.O. interview, an organization-specific questionnaire was developed to measure the influence of each department on the organization's strategic decisions and the boundary spanning activities of each department. Questionnaires were sent with a cover letter to individual department heads explaining the study and soliciting their cooperation. There were 124 usable questionnaires returned from 139 sent out, an 89 per cent response rate. A pretest of the questionnaire was conducted in two additional organizations. No changes were made in the questionnaire due to the pretest.

The questionnaire was designed to determine the influence of each department in the organization on each of a set of strategic decisions. These strategic decisions were common to each firm but were put into the 'language' of the organization after the C.E.O. interview. Figure I gives the list of strategic decisions used in financial institutions as an example.

Each respondent was asked to rate the influence of every department in his or her organization on each of the strategic decisions. This multi-rater method was used to circumvent the potential problems of bias in self-perception and importance as pointed out by Nealey and Owen (1970) and

Source of funds
Introduction/deletion of products or services
Development of new products or services
Acquisitions
Investment policies
Operating budget
Product/service pricing
Choice of markets: consumer *vs* commercial, geographic
Development and maintenance of correspondent relationships
Capital expenditures for facilities and equipment
Marketing strategies

Figure 1. Strategic decisions in financial institutions

Dearborn and Simon (1956). A list of specific strategic decisions (rather than simply the term strategic decisions) was used for two reasons. First, the list improved the construct validity of the instrument. The respondents knew what was meant by each decision. In addition, a number of decisions captures the richness of strategic decision influence present in organizations. This corresponds to the research findings that have reported differential influence of groups or individuals across a variety of decisions (Baldridge, 1971; Hinings *et al.*, 1974).

The scales used in measuring influence on strategic decisions were first used by Tannenbaum (1968) and then Hinings *et al.* (1974) and were based on a five-point, Likert-type interval scale ranging from 1 (no influence at all) to 5 (a very great deal of influence). The reliability of this scale was high. Standardized item alphas were computed for each organization and were at least $\alpha = 0.84$ or higher for each organization.

The dependent variable of interest in this study is influence on strategic decisions. Every department's influence on the strategic decisions in its organization was computed by averaging all the respondents' scores for that particular department over the range of strategic decisions. This was done by first averaging all respondents' scores for each department on each decision. Then that score on a particular decision was weighted by the respondents' perceptions of the importance of that decision to the organization. Finally, each department's influence score was averaged over all the strategic decisions of the organization [2].

It was pointed out above that a measure of boundary spanning, suitable for this research, was not available. Accordingly, an instrument was developed specifically for this research. Details of the instrument development are reported elsewhere (Jemison, 1978, 1979) but are summarized here.

An instrument composed of twenty-one questions representing actual job-related, boundary-spanning activities was developed for this study. This instrument, based on the work of Miles (1976), Leifer (1975) and Keller and Holland (1975), reflected as complete a list of actual boundary spanning activities as possible. A copy of the instrument appears in Appendix A. Respondents indicated on a 5-point, Likert-type scale the degree to which the

Table II. Boundary spanning roles, oblique solution

Strategic Boundary Roles (Factors)	Oblique factor loadings factor I	II	III	IV	V	VI	Commun-ality	Eigen Value	% Variance
Factor I – Representative								7.92	34.4%
Q4. Provide information on a formal basis to groups outside your organization that is intended to create a favourable image of your organization.	0.78	0.00	0.05	0.10	0.04	0.05	0.76		
Q5. Provide information on an informal basis to groups outside your organization that is intended to create a favourable image of your organization.	0.82	0.09	0.06	0.06	0.07	−0.01	0.79		
Q10. Make speeches to outside groups on other than specifically company business.	0.38	0.06	0.11	0.01	0.32	0.05	0.43		
Q15. Provide information on a formal basis about your organization to outsiders that will induce them to act favourably in behalf of your organization.	0.56	−0.08	0.19	0.03	0.10	0.16	0.61		
Q16. Provide information on an informal basis about your organization to outsiders that will induce them to act favourably in behalf of your organization.	0.64	−0.01	0.11	0.07	0.13	0.10	0.58		
Factor II – Physical Input Control								2.78	12.1%
Q1. Decide on the kinds of physical inputs to acquire from outside the organization, (*e.g.*, raw materials, personnel, funds, supplies, *etc.*)	0.06	0.89	0.03	−0.09	0.00	0.01	0.81		
Q2. Decide on the quality requirements for physical inputs (e.g. raw materials, personnel, funds, supplies, *etc.*)	0.01	0.89	0.11	−0.01	−0.01	−0.01	0.80		
Q3. Decide when to acquire certain physical inputs (*e.g.*, raw materials, personnel, funds, supplies)	−0.03	0.75	0.01	−0.03	0.07	−0.00	0.36		
Q6. Acquire the physical resources needed for the organization's functioning (*e.g.*, procure raw materials and supplies, negotiate a bank credit line, hire personnel).	0.00	0.47	−0.22	0.13	−0.01	0.19	0.35		
Factor III – Customer Contact and Choice								2.54	11.0%
Q11. Meet with customers and convince them to use your organization's products.	0.17	0.01	0.79	0.07	−0.04	0.01	0.77		

Q.12 Decide on the kinds of customers that your organization will pursue.	0.07	0.01	$\boxed{0.86}$	0.03	0.10	0.01	0.86		
Q17. Decide the method by which your product will be provided to your customers.	0.05	0.00	$\boxed{0.70}$	−0.06	0.03	0.00	0.53		
Factor IV – Information Acquisition – Your Needs								1.43	6.2%
Q18. Prepare formal reports for others in your organization about information that you've acquired about external factors that could influence your organization.	0.11	0.07	0.22	$\boxed{0.51}$	0.06	0.12	0.52		
Q19. Prepare informal reports for others in your organization about information that you've acquired about external factors that could influence your organization.	0.18	0.05	0.20	$\boxed{0.61}$	−0.03	0.19	0.69		
Q20. Acquire information formally from specific individuals or groups outside your organization that is needed by your department or office.	0.02	−0.05	−0.08	$\boxed{0.84}$	0.11	−0.07	0.75		
Q21. Acquire information informally from specific individuals or groups outside your organization that is needed by your department or office.	0.06	−0.04	−0.08	$\boxed{0.84}$	0.05	0.01	0.76		
Factor V – Information Acquisition – Others' Needs								1.23	5.3%
Q13. Acquire information formally from specific individuals or groups outside your own organization that is needed by a department in your organization other than your own.	0.01	0.07	0.01	0.01	$\boxed{0.80}$	0.07	0.43		
Q14. Acquire information informally from specific individuals or groups outside your organization that is needed by a department in your organization other than your own.	0.11	−0.101	−0.1	0.07	$\boxed{0.85}$	0.01	0.77		
Factor VI – Information Control								1.05	4.6%
Q7. Decide what portions of information acquired from sources outside your organization to transmit to others in your organization that will make use of it.	−0.02	0.09	0.08	0.04	0.16	$\boxed{0.76}$	0.77		
Q8. Decide when to transmit to others in your organization information acquired from outside the organization.	0.18	0.04	−0.04	−0.06	0.81	$\boxed{0.79}$			
Q9. Decide to whom information received from outside your organization should be sent.	0.07	−0.06	−0.01	0.06	0.03	$\boxed{0.83}$	0.78		

activities were 'never a part of their department's work' to 'very often a part of their department's work'. The same questions (activities) were used for all respondents. The only difference in the instrument was that in hospitals the word 'customer' was changed to 'patient' whenever it appeared. Reliability checks indicated good reliabilities for the scales developed from the instrument (see table III).

The items in appendix A were chosen to represent as broad and conceptually inclusive list of boundary spanning activities as possible. Data from the boundary spanning questions were factor analyzed to determine their dimensionality. Orthogonal and oblique factor analysis both yielded six factors explaining about 73 per cent of the variance (see table II.) The very strong factor loadings suggest factorial independence. But, since there was some inter-correlation among the oblique factors, a second order factor analysis was run on the basis of the oblique solution (Kerlinger, 1972).

This second order factor analysis revealed that there were three underlying, independent dimensions in the data explaining 70.8 per cent of the total variance (see table III.) These dimensions are the three boundary spanning roles used in this research:

Information acquisition and control
Domain determination and interface
Physical input control

The measure of a department's activity on any specific boundary spanning role (B.S.R.) was the department's factor score on the factor representing that B.S.R.

Table III. Second order factor analysis, factor loadings

	Second Order Factors		
First Order Factor from Oblique Solution	*Information acquisition and control*	*Domain determination and interface*	*Physical input control*
Representative	0.360	0.516	0.109
Customer contact	0.083	0.730	−0.063
Physical input control	0.047	−0.004	0.447
Information acquisition – own use	0.602	0.088	−0.072
Information acquisition – others' use	0.582	0.221	0.116
Information control	0.634	0.114	0.169
Eigenvalue	2.19	1.07	0.98
% Variance	36.5%	17.9%	16.4%
Cumulative % Variance	36.5%	54.4%	70.8%
Reliability of Second Order Factors			
Standardized Item Alpha	0.82	0.81	0.89

The reader may note that the eigenvalue for Factor 3 is less than the rule of thumb cut-off of 1.0. This factor was included after its logical makeup (Rummel, 1970) was considered and after performing the scree test (Cattell, 1966) and discontinuity test (Cattell, 1958). These tests showed that it was indeed a logical factor and should be included in the solution.

RESULTS

The hypotheses were tested with a series of one way analyses of variance. The dependent variables were measures of each department's influence on strategic decisions (INFLUENCE). The independent variables were the extent to which a department engaged in one of the three boundary spanning roles (B.S.R.). The results of the ANOVA procedure for the hypotheses are shown in table IV.

Hypothesis 1 is partially supported. In long-linked technologies, represented by food processors, the B.S.R. of Domain determination and interface is highly significant as hypothesized and explains 64 per cent of the variance in influence on strategic decision-making. Contrary to the hypothesis, the B.S.R. of Physical input control is not significantly associated with influence on strategic decisions in long-linked technologies.

Hypothesis 2 is partially supported. In mediating technologies, represented by financial institutions, the B.S.R. of Domain determination and interface is highly significant as hypothesized. In contrast to the hypothesis, the B.S.R. of Physical input control is not significant but that of Information acquisition and control is. Together the two significant B.S.R.s explain about 59 per cent of the variance in the dependent variable.

Hypothesis 3 is not supported. In intensive technologies, represented by general hospitals, the B.S.R. of Information acquisition and control was hypothesized to have more influence on strategic decisions. Instead, the data show that the Information B.S.R. explains less of the variance in influence on strategic decisions than do either of the other two B.S.R.s. The three B.S.R.s together explain about 61 per cent of the variance in influence on strategic decision-making.

DISCUSSION

The findings of this study can be summarized as follows:

(1) The relative influence of the three boundary spanning roles is related to the technology of the organization.
(2) Regardless of the technology, the greatest absolute amount of influence on strategic decisions is associated with the boundary spanning role of Domain determination and interface.
(3) In the sample organizations, approximately 60 per cent of the variance in influence on strategic decisions is associated with boundary spanning roles.

These findings and the unanticipated results regarding the hypotheses will now be discussed at greater length.

The results of this research suggest that the relative influence of boundary

Table IV. One-way analyses of variance: dependent variable = influence, independent variable = boundary spanning roles

(η^2)

Boundary Spanning Roles	Food Processors			Financial Institutions			General Hospitals		
	η^2	F	Sig of F	η^2	F	Sig of F	η^2	F	Sig of F
Information acquisition and control	–	–	–	0.249	2.497	0.069	0.169	2.925	0.044
Domain determination and interface	0.640	9.307	0.001	0.345	8.790	0.001	0.228	4.223	0.011
Physical input control	–	–	–	–	–	–	0.219	4.011	0.013
	0.640			0.594			0.616		

η^2 = eta^2, the percent of variance in the dependent variable explained by the independent variable.

Note: The η^2 figures are additive because of the independence of the boundary spanning roles (Rummel, 1970).

The number of respondents for each industry are:		η
	Food processors	35
	Financial institutions	46
	General hospitals	43
	Total Sample	124

roles on strategic decisions is affected by the organization's technology, that is, its internal utility-creating processes; in each technology type, the boundary spanning role of Domain determination and interface is relatively more influential than any of the other boundary roles. However, as table IV shows, its importance decreases markedly as one moves from long-linked technologies (food processors) to mediating to intensive technologies (financial institutions and general hospitals). In food processors the boundary role of Domain determination explains 64 per cent of the variance in influence on strategic decisions while in financial institutions and general hospitals the amount of variance explained decreases to about 35 per cent and 25 per cent respectively. Thus, as the technology changes, boundary spanning roles' influence on strategic decision-making changes.

It is prudent to cautiously interpret these changes in influence on strategic decisions in the absence of data that can support causal conjectures. However, it is useful and important to offer tentative explanations for these changes that have face validity. In food processors it is critically important to select an appropriate customer base and to develop and maintain a good relationship with those customers (the B.S.R. of Domain determination and interface) for two reasons – the necessity for scale economies in production runs and the competitive nature of the industry.

The investment in fixed capital by food processors and, in many cases the seasonal nature of the raw material inputs, necessitates steady, efficient production runs that produce output which must be transferred to customers (and off the company's books) as soon as possible. In addition, there are relatively fewer barriers to entry (Porter and Caves, 1977) in food processors than in financial institutions or general hospitals. Barriers to entry are lower in food processors due to relatively smaller capital requirements, the need for a less skilled labour force and fewer government regulations governing their activities. Those lower entry barriers would tend to encourage more firms to enter and thus promote more competition. Thus, one could argue that in an industry with low barriers to entry and high needs for production scale economies, the B.S.R. that selects and accommodates customers will have a relatively greater proportion of influence on the strategic decisions of firms in that industry than other B.S.R.s. The food processing industry and the B.S.R. of Domain determination and interface fulfill those conditions.

Financial institutions are economically successful if they can profitably manage the spread between their cost of funds and the amount they charge to their customers. This implies that the critical functions are customer contact and environmental scanning. A sufficient group of customers who can profitably contribute to and use the pool of funds must be maintained. In addition, economic conditions, industry trends and customer viability must be judiciously monitored. Table IV shows the importance of these activities in the relative influence of B.S.R.s of Domain determination and interface and Information acquisition and control.

In general hospitals the B.S.R. of Domain determination was surprisingly influential. This may be due to the dual need of the hospital to define a domain that meets the physicians' and community's needs as well as economically utilize the capacity of the hospital (bed space and associated diagnostic and treatment equipment). Table IV shows that the two other B.S.R.s are also significantly associated with influence on strategic decisions.

The B.S.R. of Physical input control involves deciding on the kinds of quality and timing of physical input acquisitions (*e.g.* personnel, supplies, funds) and is influential because of the tremendous importance in hospitals today of holding down costs while providing quality care. Thus the managers in these functions deal with a critical contingency for the organization because of the hospital's dual needs for low costs (*e.g.* government regulations on fee schedules) and quality inputs (who wants to go to a hospital and be treated by inferior personnel or with substandard materials or equipment?).

The other important B.S.R. in general hospitals is that of Information acquisition and control. The nature of the technology of a hospital involves bringing the patient into the hospital setting, acquiring information about the patient's physical or mental condition and then prescribing a programme of treatment. The people engaging in this B.S.R. are essential to the success of the hospital – if a patient isn't treated properly (and if the improper treatment continues for many patients) the hospital will lose its charter from society to treat patients in the future. Thus because of the importance of information in the treatment of patients, the B.S.R. of Information acquisition is influential.

It is important to note that 60 per cent of the variance in influence on strategic decisions is associated with B.S.R.s (table IV). It is also important to note that although there are slight differences among technology types, the total variance explained is still approximately 60 per cent, regardless of technology. This supports the contentions of authors who have emphasized the importance of the environment as an influence on the organization (Aldrich, 1979; Pfeffer and Salancik, 1977). It also supports the positions of those who argue for the importance of boundary spanning roles in strategic decision-making (Aldrich and Herker, 1977; Leifer and Delbecq, 1978; Thompson, 1967).

The area of organization-environment relations has a rich tradition in strategic management and organization theory. The research reported here suggests several fruitful areas of investigation that are related to influence on decision-making, organizational performance and environmental interaction. This study found that about 60 per cent of the variance in influence in strategic decisions was associated with boundary spanning activity. But, what about the determinants of the remaining 40 per cent? With what organizational, departmental or individual characteristics is this remaining influence associated? Promising areas of research include power bases, organization contextual factors (*e.g.* structure, ownership patterns) and personal characteristics of the decision-makers. Further study here will shed more light on our understanding of power in organizations, its sources and how it is used.

This study's findings also beg questions about the relationship of organizational performance and successful management of boundary spanning roles. If one believes that organizations of the future will be facing rapid change, resource scarcity and environmental complexity, then environmental interactions will become even more important to the organization's survival. Important and useful research can be undertaken that explores the relationship between organizational performance and the management of environmental interactions.

APPENDIX A

The purpose of this section is to identify the kinds of activities performed by your department or office. Please indicate in the blank beside the activity the degree to which that activity is a part of your department's or office's activities in the service of your organization. As before, all responses are completely confidential.

If the activity is *never a part of your work* enter:	1
If the activity is *seldom a part of your work* enter:	2
If the activity is *occasionally a part of your work* enter:	3
If the activity is *frequently a part of your work* enter:	4
If the activity is *very often a part of your work* enter:	5

1. Decide on the kinds of physical inputs to acquire from outside the organization (*e.g.* funds, personnel, supplies). ______
2. Decide on the quality requirements for physical inputs (*e.g.* funds, personnel supplies) to be acquired from outside the organization. ______
3. Decide when to acquire certain physical inputs (*e.g.* funds, personnel, supplies) from outside the organization. ______
4. Provide information on a *formal* basis to groups outside your organization that is intended to create a favourable image of your organization. ______
5. Provide information on an *informal* basis to groups outside your organization that is intended to create a favourable image of your organization. ______
6. Acquire the physical resources needed for the organization's functioning (*e.g.* negotiate a bank credit line, hire personnel, procure supplies). ______
7. Decide what portions of information acquired from sources outside your organization to transmit to others in your organization that will make use of it. ______
8. Decide when to transmit to others in your organization information acquired from outside the organization. ______
9. Decide to whom information received from outside your organization should be sent. ______
10. Make speeches to outside groups on other than specifically company business. ______

11. Meet with customers and convince them to use your organization's products. ________
12. Decide on the kinds of customers that your organization will pursue. ________
13. Acquire information *formally* from specific individuals or groups outside your organization that is needed by a department in your organization other than your own. ________
14. Acquire information *informally* from specific individuals or groups outside your organization that is needed by a department in your organization other than your own. ________
15. Provide information on a *formal* basis about your organization to outsiders that will induce them to act favourably on behalf of your organization. ________
16. Provide information on an *informal* basis about your organization to outsiders that will induce them to act favourably on behalf of your organization. ________
17. Decide the method by which your product will be provided to your customers. ________
18. Prepare *formal* reports for others in your organization about information that you've acquired about external factors that could influence your organization. ________
19. Prepare *informal* reports for others in your organization about information that you've acquired about external factors that could influence your organization. ________
20. Acquire information *formally* from specific individuals or groups outside your organization that is needed by your department or office. ________
21. Acquire information *informally* from specific individuals or groups outside your organization that is needed by your department or office. ________

NOTES

[1] I would like to thank *Journal of Management Studies* referees for their constructive comments.

[2] where:

$$\text{INFLUENCE}_m = \frac{\sum_{k=1}^{p} \sum_{j=1}^{n} (\text{DI}_{mkj}) \cdot (\text{W}_{kj})}{(n)\ (p)}$$

INFLUENCE_m = weighted mean influence of department m on k decisions as scored by n respondents

DI_{mjk} = influence of department m on decision k as scored by respondent j

W_{kj} = importance weight of decision k as scored by respondent j.

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